

KUNAL KUMAR SINGH

H85 near Axis Bank, Noida 201303

Phone: 9616714403 Email:Ks286823@gmail.com

GitHub: <https://github.com/WhozKunal> | LinkedIn: <https://www.linkedin.com/in/itskunal>

PROFESSIONAL SUMMARY

Results-driven Python Developer (AI/ML) with 4+ years of experience in designing, developing, and deploying AI-driven applications. Expertise in machine learning, NLP, computer vision, and LLM-powered applications. Strong experience in Tensor Flow, PyTorch, LangChain, Hugging Face, and OpenAI APIs. Proven ability to optimize large-scale data processing, model deployment, and cloud-based AI solutions.

Python Developer AI/ML	Upsala Detsol And It Services Private Limited	Dec 2023 –Working - Delhi
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- Develop AI-driven applications that involve natural language understanding, such as chatbots, virtual assistants, and voice recognition systems.
- Create NLP models using frameworks like TensorFlow, PyTorch, or Hugging Face Transformers.
- Collect and pre-process datasets for training NLP models, including tasks like tokenization, stemming, lemmatization, and part-of-speech tagging.
- Integrate NLP models into applications, ensuring scalability, reliability, and security.
- Collaborate with frontend and backend developers to integrate NLP functionality into web or mobile applications.

Python Developer AI/ML	Aditya Intellectual	Aug 2020 - Sep 2023 - Surat
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- **Model Development and Optimization:** Designed, developed, and optimized machine learning models using PyTorch and TensorFlow. Applied supervised learning techniques to build models for classification, regression, and structured prediction tasks.
- **Data Management:** Handled and preprocessed large-scale datasets, including data cleaning, augmentation, and transformation. Utilized visualization tools to interpret data and communicate insights effectively.
- **Algorithm Implementation:** Implemented and fine-tuned supervised learning algorithms such as neural networks, decision trees, and support vector machines. Developed custom loss functions and optimization strategies to enhance model performance.
- **Deployment and Integration:** Deployed machine learning models to production environments using TensorFlow Serving and PyTorch's TorchServe. Collaborated with software engineers to integrate models with APIs and existing systems.
- **Performance Monitoring:** Monitored and evaluated model performance using metrics like accuracy, precision, recall, and F1 score. Identified and addressed issues related to model drift, overfitting, and underfitting.
- **Collaboration:** Worked closely with cross-functional teams, including data scientists, engineers, and product managers, to understand requirements and deliver actionable insights. Participated in code reviews and knowledge-sharing sessions.

- **Research and Innovation:** Conducted research on the latest advancements in supervised learning and neural network architectures. Implemented innovative approaches to improve model capabilities and accuracy.

EDUCATION

Gujarat, Surat	Veer Narmad South Gujarat University	Feb 2018 - May 2021
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- B.C.A in Computer Science Engineering with Minor in Mathematics May 2021. In-major GPA: 5.6
- Graduate Coursework: Software Foundations; Computer Architecture; Algorithms; Artificial Intelligence; Comparison of Learning Algorithms; Computational Theory.
- Undergraduate Coursework: Operating Systems; Databases; Algorithms; Programming Languages; Comp. Architecture; Engineering Entrepreneurship; Calculus III.

Projects

Access Control Surveillance System:

- This project leverages the Open VINO framework to implement a machine learning-based surveillance system for face identification.
- Using transfer learning, I trained a custom model to accurately recognize faces for access control purposes.
- The system integrates advanced computer vision techniques to enhance security in restricted areas.

Key Features:

- Face identification for real-time access control.
- Model training using OpenVINO's optimized deep learning framework for high performance on edge devices.
- Incorporates transfer learning to fine-tune a pre-trained model, ensuring efficiency and accuracy with limited data.

VisionAI - Automated Image-Based Data Extraction for Land & Fiber Networks

- Developed **VisionAI**, an AI-powered system that automates data extraction from satellite and infrastructure images to assist in land base creation and fiber network analysis. The system processes images to extract critical geographical and structural features, enabling automated decision-making for permit approvals and network planning.

Real-Time Object Detection for Autonomous Vehicles:

- **Description:** Create an object detection system to identify pedestrians, vehicles, and road signs in real-time video feeds.
Key Skills: YOLO, OpenCV, TensorFlow/PyTorch.

AI-Powered Knowledge Base Chatbot:

- Use LangChain and ChromaDB to build a chatbot that answers questions based on specific documents, such as company policies, product manuals, or research papers.
- Integrate AI agents to perform multi-step reasoning and retrieve information from structured and unstructured data.
- **Tech Stack:** LangChain, ChromaDB, OpenAI/LLM APIs, Streamlit.

Autonomous AI Agent for Business Automation:

- Create an AI agent that can autonomously perform tasks like market research, competitive analysis, or generating marketing plans by querying APIs and acting iteratively.
- Integrate tools like LangChain, Pinecone/ChromaDB, and AI Functions.
- **Tech Stack:** LangChain, GPT-4, Pinecone, Python.

Intelligent Chatbot with Custom Dataset:

- **Description:** Build a chatbot using Mistral LLM or similar, trained on a domain-specific dataset (e.g., recruitment, healthcare, or legal).
Key Skills: NLP, fine-tuning pre-trained models, APIs.

E-commerce website with Django. Created a fully functional e-commerce website Django web framework.

ADDITIONAL EXPERIENCE AND AWARDS

- **AWS Certification Course:** Completed AWS course from Guvi learn Aws deployment
- **Python Developer Course AI/ML:** Completed a Python AI/ML course for Ducat University

Languages and Technologies

- **Programming Languages:**
Python (Advanced), C/C++, JavaScript/TypeScript, HTML5/CSS3
- **Machine Learning & AI:**
Frameworks: TensorFlow, PyTorch, scikit-learn, JAX
Large Language Models: LangChain, LlamaIndex, OpenAI API
Vector Databases: ChromaDB, Pinecone, Weaviate, Milvus
Model Deployment: BentoML, MLflow, Gradio
Inference Engines: ONNX Runtime, TensorRT
Model Serving: Ollama, groq, Hugging Face Inference API
Triton Inference Server, NVIDIA DeepStream,

- **Natural Language Processing:**
Transformers (Hugging Face)
spaCy, NLTK
Whisper (Speech Recognition)
Pyannote (Speaker Diarization)
word2vec, fastText, BERT variants
- **Computer Vision:**
OpenCV
YOLO (Object Detection)
MediaPipe
DeepFace (Face Recognition)
Detectron2
Albumentations (Image Augmentation)
- **Web Development:**
Backend: Django, FastAPI, Flask
API Design: REST, GraphQL
Testing: Jest, Pytest
- **Database Technologies:**
Relational: MySQL
ORM: SQLAlchemy, Django ORM
- **Cloud & DevOps:**
Platforms: AWS, GCP, Azure
Containerization: Docker, Kubernetes
CI/CD: Jenkins, GitHub Actions, GitLab CI
Infrastructure as Code: Terraform, Ansible
Monitoring: Prometheus, Grafana
- **Image Processing:** Tesseract OCR, Pillow, YOLO, Mask R-CNN
- **Development Tools:**
Version Control: Git, GitHub, GitLab
IDE: VSCode, PyCharm, Jupyter
Package Management: pip, conda, poetry
- **Geospatial Analysis:** Google Maps API, GIS tools